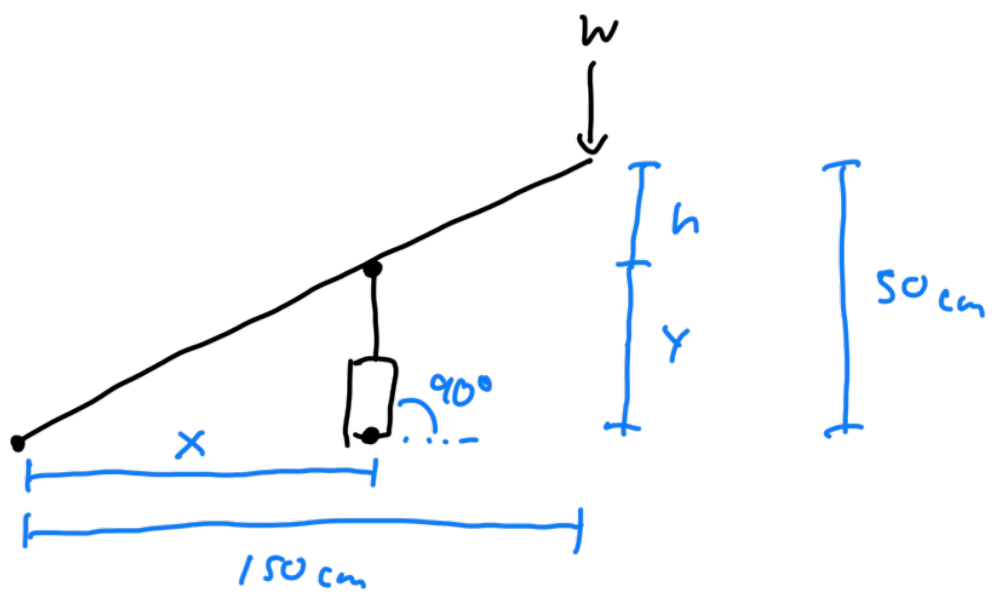
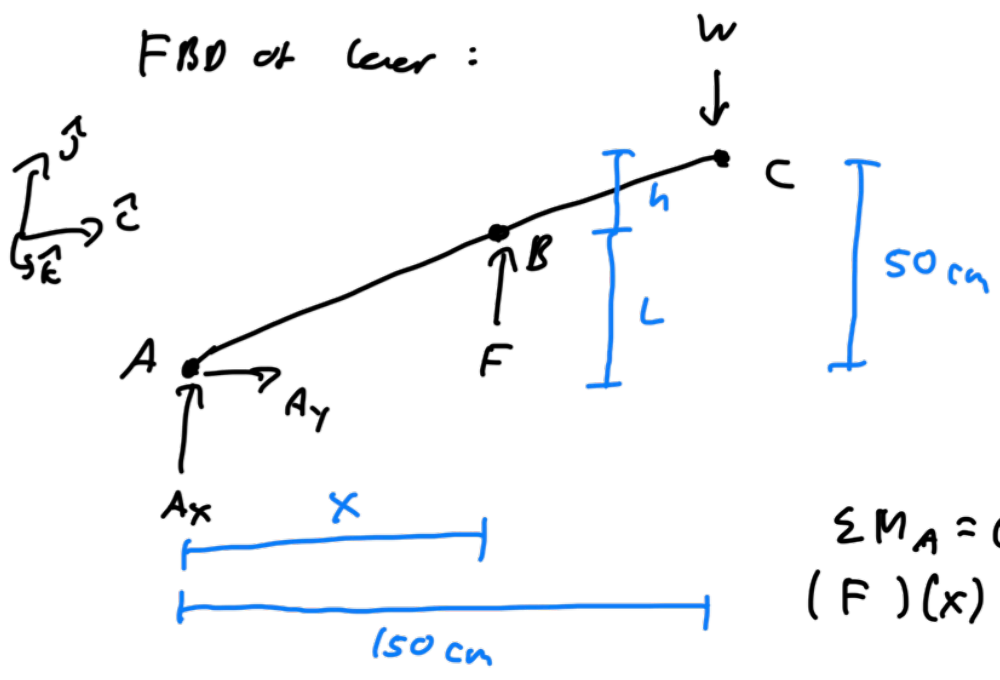


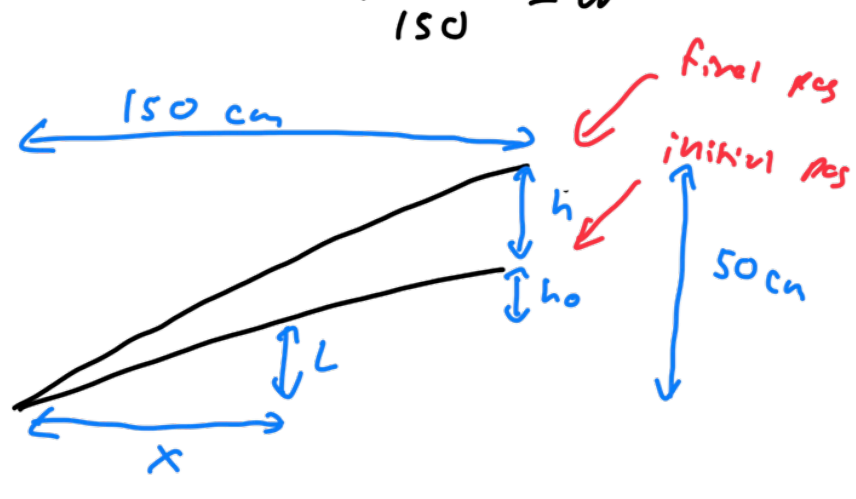
The goal is to lift the maximum weight to the maximum height.

To find the max W for a given actuator with max force F, I use a FBD of the lever.

FBD of lever:



$$\Sigma M_A = 0$$
$$(F)(x) - (W)(150\text{ cm}) = 0$$
$$\frac{Fx}{150} = W$$



$h = 50\text{cm} - L - h_0$
$h = 50 - L - (\frac{1}{3}(150 - x))$
$h = -L + \frac{1}{3}x$

$h_0 = \frac{50\text{ cm}}{150\text{ cm}} \cdot (150\text{ cm} - x)$
$h_0 = \frac{1}{3}(150 - x)$

Maximize the quantity hW

(lifted height x weight is metric for quality)

maximize $(-L + \frac{1}{3}x)(\frac{Fx}{150})$

$$f(F, x, L) = \frac{-1}{150}FxL + \frac{1}{450}Fx^2$$

where F = max force of actuator
x = distance along lever
L = min length of actuator

X should be as large as possible (150 cm) so we have

$$f(F, L) = -FL + 50F$$

constrained to $L < 50$

Product manual listing length as "BB + stroke + P", chose min values for stroke and P

IMA 55: L = 49.24 cm, F = 35.81 kN
f(35.81, 49.24) = 27.22

max peak thrust value

Final choice of actuator

IMA 44: L = 41.49 cm, F = 18.46 kN
f = 157.09

IMA 33 3 stack: L = 36.91 cm, F = 11.34 kN
f = 148.44

IMA 33 1 stack: L = 29.29 cm, F = 6.01 kN
f = 124.47

IMA 22 3 stack: L = 32.22 cm, F = 1.45 kN
f = 25.78

IMA 22 1 stack: L = 24.60 cm, F = 1.45 kN
f = 36.83